



Technological University Dublin (Tallaght Campus)

School of Enterprise Computing and Digital Transformation

Module: WADT H1002

Web User Experience and Database Design

Supplemental CA in August/September 2026

Lecturer: Inthraporn Aranyanak (Joop)

Lecturer: Michael A. Murphy

Onsite QUIZ (30%)

The onsite quiz will take place at 10:00 AM on Wednesday, 2 September 2026, in Lab 229.

The quiz consists of 30 questions, and you will have 25 minutes to complete it. There will be 15 questions on Web UX and 15 questions on Database Design, covering topics from both Semester 1 and Semester 2. The quiz is a closed book.

Please arrive 10 minutes early and ensure that you are able to log in to Brightspace before the quiz begins.

Individual Project (70%)

Developing a Web Application Integrating Frontend and Database

Overview: Students are required to design and develop an individual database-driven web application. You are encouraged to choose a project theme based on your own interests, taking inspiration from the course material. Possible application domains include e-commerce, social media, entertainment, education, event management, fitness, booking systems, inventory management, and similar areas. There is no requirement for users to log in to the system, although your application will need login credentials to access the database, but that is a once-off. The project should consist of multiple related web pages that manage data stored in a relational database. The application must implement full CRUD functionality:

- Create new records
- Retrieve and display records
- Update existing records
- Delete records

If you have already completed a project as part of the Semester 2 team project, you must not reuse or submit the same project for this assessment. Instead, you are required to develop a new database-driven web application from scratch. This requirement ensures that all students have an equal opportunity to demonstrate their individual design and development skills.

The presentation date (Mandatory): Wednesday, 2 September 2026, in Lab 229 after the quiz.

Learning Outcomes:

- Design and develop a database-driven web application
- Implement core CRUD functionality (create, retrieve, update, and delete data) using multiple related database tables
- Develop and integrate frontend web pages with backend database access
- Present and explain technical design decisions, and implementation details

Project Requirement:

Web Application

- A Home page plus at least one additional webpage (minimum of two webpages in total).
- Forms for entering and updating data.
- Tables or other suitable components for displaying database records.
- Full CRUD functionality across the application.

Database

- A minimum of **two related database tables**.
- All database tables must be created in **Supabase using SQL scripts only**.
- The SQL setup script must **not** be generated automatically using the Supabase Table Editor or export tools.
- Your SQL definitions should demonstrate an understanding of database design by implementing appropriate **integrity constraints**, where applicable, such as:
 - Primary Keys
 - Foreign Keys
 - NOT NULL constraints
 - CHECK constraints

Version Control

- Your project must be maintained in an individual GitHub repository.
- The repository must include:
 - Frontend source code
 - Backend/database connection code
 - A SQL database setup script that can recreate the database schema.

Grading Rubric: (70%)

- Presentation onsite (Mandatory) – 20%
- Website Design & Functionality & Accessibility - 20%
- Database Definition - 20%
- Upload your codes on GitHub (before Wednesday, 2 September 2026)- 10%

Submission Checklist:

- Please upload your HTML, CSS, Javascript, images files, Entity Relationship Diagram (ERD), and SQL scripts on your Github. Please invite me to your Github link using this account. (inthraporn.aranyanak@tudublin.ie)
- Please invite Inthraporn and Michael to access tables in supabase. (inthraporn.aranyanak@tudublin.ie) and (MichaelA.Murphy@tudublin.ie)

- Please submit your GitHub link & a link to Supabase on Brightspace in folder: Supplemental CA in August/September 2026.